# Linux Web Server Deployment and Automated System Monitoring

**GitHub Project Link:** https://github.com/Neeraj13579/Linux-Web-Server-Deployment-and-Automated-System-Monitoring/

**Project Overview**

In this capstone, you will set up a production-style Linux web server from scratch. You will install and configure Nginx, manage packages, configure the firewall, monitor logs, and write a shell script that automatically checks server health. This mirrors what a junior Linux engineer does when setting up and maintaining a web server in a real company.

**Topics Covered:** Nginx | Package Management (YUM/APT) | Firewall Management | Log Monitoring | rsyslog | Shell Scripting | Network Commands | SSH | systemctl | crontab | grep | tail | curl | wget

**Environment:** AWS EC2 t2.micro (Amazon Linux 2023) — Free Tier **OR** Local VirtualBox VM (CentOS/Ubuntu)

### Linux Web Server Setup — Nginx, Firewall, Package Management & Log Monitoring

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**Project Architecture**

Linux Web Server (EC2 or VM)
├── Nginx Web Server
│ ├── Default site (port 80)
│ └── Custom HTML page
├── Firewall (firewalld)
│ ├── Allow HTTP (80)
│ ├── Allow SSH (22)
│ └── Block everything else
├── Log Monitoring
│ ├── /var/log/nginx/access.log
│ └── /var/log/nginx/error.log
└── Shell Script
└── server_health_check.sh (auto-runs every 5 min via cron)

**Setup — Launch Your Environment**

**Option A — AWS EC2 (Free Tier):**

- Launch **t2.micro Amazon Linux 2023** instance
- Security Group: allow SSH (22) from My IP | HTTP (80) from Anywhere
- Connect: ssh -i "your-key.pem" ec2-user@<public-ip>

**Option B — Local VirtualBox VM:**

- Start your CentOS or Ubuntu VM

- Open terminal
- Switch to root: sudo su -

## Phase 1 — Package Management & Nginx Installation

**Step 1: System update and package management basics**

***Commands:***

```
sudo su -

# Check current packages installed
rpm -qa | wc -l          # CentOS/Amazon Linux
# dpkg --list | wc -l    # Ubuntu

# Update all packages
yum update -y            # CentOS/Amazon Linux
# apt update && apt upgrade -y   # Ubuntu

# Search for a package
yum search nginx
yum info nginx

# Install useful tools
yum install -y wget curl tree net-tools bind-utils

# Verify installation
which wget
which curl
which tree

# Check package info
rpm -qi curl
rpm -ql curl | head -10

# List recently installed packages
rpm -qa --last | head -10
```

**Output:**

```
root@ip-192-168-1-235:~
Microsoft Windows [Version 10.0.26200.9711]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NeerajKumar>cd Downloads

C:\Users\NeerajKumar\Downloads>dir
 Volume in drive C is OS
 Volume Serial Number is 7407-60EF

 Directory of C:\Users\NeerajKumar\Downloads

11/07/2026  06:01 PM    <DIR>          .
11/09/2026  01:21 AM    <DIR>          ..
11/02/2026  05:00 PM            22,376 063f8190-7aee-4e41-9a30-6c30d00902b1 (1).pdf
11/04/2026  01:06 PM            22,376 063f8190-7aee-4e41-9a30-6c30d00902b1 (2).pdf
11/05/2026  10:22 AM            22,376 063f8190-7aee-4e41-9a30-6c30d00902b1 (3).pdf
09/24/2026  04:38 PM            22,376 063f8190-7aee-4e41-9a30-6c30d00902b1.pdf
10/19/2026  03:31 PM           140,219 1Docs.pdf
09/24/2026  04:37 PM            22,632 41f57207-2fb9-4c31-b399-460a47de6cdb.pdf
10/15/2026  05:32 PM         5,948,752 Agreement List.pdf
11/02/2026  03:21 PM             4,738 ALLCOMMERCIAL.INFRA@jsw.in_exportGroupMembers_2026-10-14 (1).csv
11/04/2026  10:40 AM           520,798 ALLCOMMERCIAL.INFRA@jsw.in_exportGroupMembers_2026-10-14.csv
10/19/2026  03:31 PM         1,397,241 All_AB1_Docs.pdf
10/19/2026  03:29 PM           211,472 All_AB_Both_Docs.pdf
09/02/2026  03:22 PM             1,678 Amazon_key.pem
09/02/2026  03:31 PM             1,462 Ama_Key.ppk
10/08/2026  05:31 PM        17,248,616 Asset Inv Phase 2 locations (1).xlsx
10/08/2026  01:22 PM        17,426,005 Asset Inv Phase 2 locations.xlsx
09/23/2026  03:41 PM            41,260 CEP_Mapping_IS_Phase2.4 1 6.xlsx
10/08/2026  05:31 PM             5,810 Client Details_2026-10-08-11-22-08.csv
09/15/2026  01:03 PM             7,503 Cloud_Career_Roadmap.md
09/20/2026  11:53 AM           152,707 COMMA SEPARATOR TOOL 4.xlsx
10/26/2026  11:19 AM         1,919,654 Commvault agent installation SOP for MAC 1.docx
09/23/2026  03:41 PM            58,450 Copilot tracker - Inventory cleanup.xlsx
09/16/2026  04:04 PM         2,139,566 CV_Mapped_to_Location_List.pdf
09/15/2026  10:27 AM               255 Days Aging Formula.txt
10/23/2026  01:15 PM         8,328,255 DevicesWithInventory_0d7b42ad-40bf-4d66-8e56-c09ee439ab21.zip
```

```
root@ip-192-168-1-235:~
C:\Users\NeerajKumar\Downloads>ssh -i "practice_key.pem" ec2-user@13.235.77.190
The authenticity of host '13.235.77.190 (13.235.77.190)' can't be established.
ED25519 key fingerprint is SHA256:r0xRPNBaTAhlrFeyqLQNkZhJtQAO59oBkcIKyM46QLY.
This host key is known by the following other names/addresses:
    C:\Users\NeerajKumar/.ssh/known_hosts:8: 65.0.45.231
    C:\Users\NeerajKumar/.ssh/known_hosts:10: 15.206.174.223
    C:\Users\NeerajKumar/.ssh/known_hosts:11: 13.201.83.221
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.235.77.190' (ED25519) to the list of known hosts.

A newer release of "Amazon Linux" is available.
  Version 2023.12.20261012:
  Version 2023.12.20261013:
  Version 2023.12.20261014:
  Version 2023.12.20261016:
  Version 2023.12.20261021:
  Version 2023.12.20261026:
  Version 2023.12.20261028:
  Version 2023.12.20261029:
  Version 2023.12.20261102:
Run "/usr/bin/dnf check-release-update" for full release and version update info
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Oct 29 08:03:25 2026 from 152.58.46.241
[ec2-user@ip-192-168-1-235 ~]$ sudo su -
Last login: Thu Oct 29 08:03:29 UTC 2026 on pts/1
[root@ip-192-168-1-235 ~]# rpm -qa | wc -l
510
[root@ip-192-168-1-235 ~]# yum update -y
```

```
[root@ip-192-168-1-235 ~]# yum update -y
Last metadata expiration check: 0:05:33 ago on Sun Nov  8 20:08:48 2026.
================================================================================================================================
WARNING:
  A newer release of "Amazon Linux" is available.

  Available Versions:

  Version 2023.12.20261012:
    Run the following command to upgrade to 2023.12.20261012:
```

```
Nothing to do.
Complete!
[root@ip-192-168-1-235 ~]# yum search nginx
Last metadata expiration check: 0:05:53 ago on Sun Nov  8 20:08:48 2026.
================================================ Name Exactly Matched: nginx ================================================
nginx.x86_64 : A high performance web server and reverse proxy server
================================================ Name & Summary Matched: nginx ================================================
collectd-nginx.x86_64 : Nginx plugin for collectd
nginx-all-modules.noarch : A meta package that installs all available Nginx modules
nginx-core.x86_64 : nginx minimal core
nginx-filesystem.noarch : The basic directory layout for the Nginx server
nginx-mimetypes.noarch : MIME type mappings for nginx
nginx-mod-devel.x86_64 : Nginx module development files
nginx-mod-http-image-filter.x86_64 : Nginx HTTP image filter module
nginx-mod-http-perl.x86_64 : Nginx HTTP perl module
nginx-mod-http-xslt-filter.x86_64 : Nginx XSLT module
nginx-mod-mail.x86_64 : Nginx mail modules
nginx-mod-njs.x86_64 : NGINX JavaScript (njs) dynamic modules for nginx
nginx-mod-stream.x86_64 : Nginx stream modules
python3-certbot-nginx.noarch : The nginx plugin for certbot
================================================ Name Matched: nginx ================================================
nginx-mod-headers-more.x86_64 : This module allows adding, setting, or clearing specified input/output headers
[root@ip-192-168-1-235 ~]# yum info nginx
Last metadata expiration check: 0:06:04 ago on Sun Nov  8 20:08:48 2026.
Available Packages
Name         : nginx
Epoch        : 1
Version      : 1.30.5
Release      : 1.amzn2023.0.1
Architecture : x86_64
Size         : 34 k
Source       : nginx-1.30.5-1.amzn2023.0.1.src.rpm
Repository   : amazonlinux
Summary      : A high performance web server and reverse proxy server
URL          : https://nginx.org
License      : BSD-2-Clause
Description  : Nginx is a web server and a reverse proxy server for HTTP, SMTP, POP3 and
```

```
Description  : Nginx is a web server and a reverse proxy server for HTTP, SMTP, POP3 and
             : IMAP protocols, with a strong focus on high concurrency, performance and low
             : memory usage.

[root@ip-192-168-1-235 ~]# yum install -y wget curl tree net-tools bind-utils
Last metadata expiration check: 0:06:44 ago on Sun Nov  8 20:08:48 2026.
Package wget-1.21.3-1.amzn2023.0.5.x86_64 is already installed.
Package tree-1.8.0-6.amzn2023.0.2.x86_64 is already installed.
Package net-tools-2.0-0.59.20160912git.amzn2023.0.3.x86_64 is already installed.
Package bind-utils-32:9.18.50-1.amzn2023.0.3.x86_64 is already installed.
Error:
 Problem: problem with installed package curl-minimal-8.21.0-5.amzn2023.0.2.x86_64
  - package curl-minimal-8.21.0-5.amzn2023.0.2.x86_64 from @System conflicts with curl provided by curl-7.87.0-2.amzn2023.0.2.x86_64
from amazonlinux
  - package curl-minimal-7.87.0-2.amzn2023.0.2.x86_64 from amazonlinux conflicts with curl provided by curl-7.87.0-2.amzn2023.0.2.x86
_64 from amazonlinux
  - package curl-minimal-7.88.0-1.amzn2023.0.1.x86_64 from amazonlinux conflicts with curl provided by curl-7.87.0-2.amzn2023.0.2.x86
_64 from amazonlinux
  - package curl-minimal-7.88.1-1.amzn2023.0.1.x86_64 from amazonlinux conflicts with curl provided by curl-7.87.0-2.amzn2023.0.2.x86
_64 from amazonlinux
  - package curl-minimal-8.0.1-1.amzn2023.0.1.x86_64 from amazonlinux conflicts with curl provided by curl-7.87.0-2.amzn2023.0.2.x86_
64 from amazonlinux
  - package curl-minimal-8.0.1-1.amzn2023.x86_64 from amazonlinux conflicts with curl provided by curl-7.87.0-2.amzn2023.0.2.x86_64 f
rom amazonlinux
  - package curl-minimal-8.11.1-4.amzn2023.0.1.x86_64 from amazonlinux conflicts with curl provided by curl-7.87.0-2.amzn2023.0.2.x86
_64 from amazonlinux
  - package curl-minimal-8.11.1-4.amzn2023.0.3.x86_64 from amazonlinux conflicts with curl provided by curl-7.87.0-2.amzn2023.0.2.x86
_64 from amazonlinux
  - package curl-minimal-8.15.0-4.amzn2023.0.1.x86_64 from amazonlinux conflicts with curl provided by curl-7.87.0-2.amzn2023.0.2.x86
_64 from amazonlinux
  - package curl-minimal-8.17.0-1.amzn2023.0.1.x86_64 from amazonlinux conflicts with curl provided by curl-7.87.0-2.amzn2023.0.2.x86
_64 from amazonlinux
  - package curl-minimal-8.17.0-1.amzn2023.0.2.x86_64 from amazonlinux conflicts with curl provided by curl-7.87.0-2.amzn2023.0.2.x86
_64 from amazonlinux
  - package curl-minimal-8.17.0-1.amzn2023.0.3.x86_64 from amazonlinux conflicts with curl provided by curl-7.87.0-2.amzn2023.0.2.x86
_64 from amazonlinux
  - package curl-minimal-8.2.1-1.amzn2023.0.2.x86_64 from amazonlinux conflicts with curl provided by curl-7.87.0-2.amzn2023.0.2.x86_
```

```
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# which wget
/usr/bin/wget
[root@ip-192-168-1-235 ~]# which curl
/usr/bin/curl
[root@ip-192-168-1-235 ~]# which tree
/usr/bin/tree
```

```
[root@ip-192-168-1-235 ~]# rpm -qi curl
Name        : curl
Version     : 8.21.0
Release     : 5.amzn2023.0.2
Architecture: x86_64
Install Date: Sun Nov  8 20:20:12 2026
Group       : Unspecified
Size        : 533579
License     : curl
Signature   : RSA/SHA512, Tue Sep 22 20:06:58 2026, Key ID e951904ad832c631
```

```
Source RPM  : curl-8.21.0-5.amzn2023.0.2.src.rpm
Build Date  : Tue Sep 22 20:01:25 2026
Build Host  : koji.amazonlinux
Packager    : Amazon Linux
Vendor      : Amazon Linux
URL         : https://curl.se/
Summary     : A utility for getting files from remote servers (FTP, HTTP, and others)
Description :
curl is a command line tool for transferring data with URL syntax, supporting
FTP, FTPS, HTTP, HTTPS, SCP, SFTP, TFTP, TELNET, DICT, LDAP, LDAPS, FILE, IMAP,
SMTP, POP3 and RTSP.  curl supports SSL certificates, HTTP POST, HTTP PUT, FTP
uploading, HTTP form based upload, proxies, cookies, user+password
authentication (Basic, Digest, NTLM, Negotiate, kerberos...), file transfer
resume, proxy tunneling and a busload of other useful tricks.
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# rpm -qi curl | head -10
Name        : curl
Version     : 8.21.0
Release     : 5.amzn2023.0.2
Architecture: x86_64
Install Date: Sun Nov  8 20:20:12 2026
Group       : Unspecified
Size        : 533579
License     : curl
Signature   : RSA/SHA512, Tue Sep 22 20:06:58 2026, Key ID e951904ad832c631
Source RPM  : curl-8.21.0-5.amzn2023.0.2.src.rpm
[root@ip-192-168-1-235 ~]#
```

```
[root@ip-192-168-1-235 ~]# rpm -qa --last | head -10
curl-8.21.0-5.amzn2023.0.2.x86_64             Sun Nov  8 20:20:12 2026
cronie-anacron-1.5.7-1.amzn2023.0.2.x86_64    Thu Oct 29 07:54:37 2026
cronie-1.5.7-1.amzn2023.0.2.x86_64            Thu Oct 29 07:54:37 2026
mod_lua-2.4.68-1.amzn2023.0.1.x86_64          Thu Oct 29 07:41:46 2026
mod_http2-2.0.42-1.amzn2023.0.1.x86_64        Thu Oct 29 07:41:46 2026
mailcap-2.1.49-3.amzn2023.0.3.noarch          Thu Oct 29 07:41:46 2026
libbrotli-1.1.0-1.amzn2023.0.1.x86_64         Thu Oct 29 07:41:46 2026
httpd-tools-2.4.68-1.amzn2023.0.1.x86_64      Thu Oct 29 07:41:46 2026
httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch Thu Oct 29 07:41:46 2026
httpd-core-2.4.68-1.amzn2023.0.1.x86_64       Thu Oct 29 07:41:46 2026
[root@ip-192-168-1-235 ~]#
```

**Step 2: Install and configure Nginx**

***Comands:***

```
# Install Nginx
yum install -y nginx      # Amazon Linux/CentOS
# apt install -y nginx    # Ubuntu

# Verify installation
nginx -v
which nginx
rpm -ql nginx | head -20

# Start and enable Nginx
systemctl start nginx
systemctl enable nginx
systemctl status nginx

# Check Nginx is listening on port 80
ss -tlnp | grep :80
netstat -tlnp | grep nginx

# Test locally
curl http://localhost
curl -I http://localhost   # headers only
```

**Output:**

```
[root@ip-192-168-1-235 ~]# yum install -y nginx
Last metadata expiration check: 0:29:37 ago on Sun Nov  8 20:08:48 2026.
Dependencies resolved.
================================================================================================================================================
 Package                          Architecture              Version                                  Repository                            Size
================================================================================================================================================
Installing:
 nginx                            x86_64                    1:1.30.5-1.amzn2023.0.1                  amazonlinux                           34 k
Installing dependencies:
 gperftools-libs                  x86_64                    2.9.1-1.amzn2023.0.3                     amazonlinux                          308 k
 libunwind                        x86_64                    1.4.0-5.amzn2023.0.3                     amazonlinux                           66 k
 nginx-core                       x86_64                    1:1.30.5-1.amzn2023.0.1                  amazonlinux                          711 k
 nginx-filesystem                 noarch                    1:1.30.5-1.amzn2023.0.1                  amazonlinux                           10 k
 nginx-mimetypes                  noarch                    2.1.49-3.amzn2023.0.3                    amazonlinux                           21 k

Transaction Summary
================================================================================================================================================
Install  6 Packages

Total download size: 1.1 M
Installed size: 3.7 M
Downloading Packages:
(1/6): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm                                                  9.0 MB/s | 308 kB     00:00
(2/6): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm                                                        1.8 MB/s |  66 kB     00:00
(3/6): nginx-1.30.5-1.amzn2023.0.1.x86_64.rpm                                                           876 kB/s |  34 kB     00:00
(4/6): nginx-filesystem-1.30.5-1.amzn2023.0.1.noarch.rpm                                                403 kB/s |  10 kB     00:00
(5/6): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm                                                 811 kB/s |  21 kB     00:00
(6/6): nginx-core-1.30.5-1.amzn2023.0.1.x86_64.rpm                                                       16 MB/s | 711 kB     00:00
------------------------------------------------------------------------------------------------------------------------------------------------
Total                                                                                                   8.3 MB/s | 1.1 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                                      1/1
  Running scriptlet: nginx-filesystem-1:1.30.5-1.amzn2023.0.1.noarch                                                                      1/6
  Installing       : nginx-filesystem-1:1.30.5-1.amzn2023.0.1.noarch                                                                      1/6
  Installing       : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch                                                                         2/6
  Installing       : libunwind-1.4.0-5.amzn2023.0.3.x86_64                                                                                3/6
```

```
Installed:
  gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64       libunwind-1.4.0-5.amzn2023.0.3.x86_64          nginx-1:1.30.5-1.amzn2023.0.1.x86_64
  nginx-core-1:1.30.5-1.amzn2023.0.1.x86_64         nginx-filesystem-1:1.30.5-1.amzn2023.0.1.noarch nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
[root@ip-192-168-1-235 ~]# nginx -v
nginx version: nginx/1.30.5
[root@ip-192-168-1-235 ~]# which nginx
/usr/sbin/nginx
[root@ip-192-168-1-235 ~]# rpm -ql nginx | head -20
/etc/logrotate.d/nginx
/usr/bin/nginx-upgrade
/usr/lib/systemd/system/nginx.service
/usr/libexec/nginx-ssl-pass-dialog
/usr/share/man/man3/nginx.3pm.gz
/usr/share/man/man8/nginx-upgrade.8.gz
/usr/share/man/man8/nginx.8.gz
/usr/share/nginx/html/404.html
/usr/share/nginx/html/50x.html
/usr/share/nginx/html/icons
/usr/share/nginx/html/icons/poweredby.png
/usr/share/nginx/html/index.html
/usr/share/nginx/html/nginx-logo.png
/usr/share/nginx/html/poweredby.png
/usr/share/vim/vimfiles/ftdetect/nginx.vim
/usr/share/vim/vimfiles/ftplugin/nginx.vim
/usr/share/vim/vimfiles/indent/nginx.vim
/usr/share/vim/vimfiles/syntax/nginx.vim
```

```
[root@ip-192-168-1-235 ~]# systemctl start nginx
[root@ip-192-168-1-235 ~]# systemctl enable nginx
[root@ip-192-168-1-235 ~]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
     Active: active (running) since Sun 2026-11-08 20:44:06 UTC; 15s ago
   Main PID: 5400 (nginx)
      Tasks: 2 (limit: 1113)
     Memory: 2.7M
        CPU: 39ms
     CGroup: /system.slice/nginx.service
             ├─5400 "nginx: master process /usr/sbin/nginx"
             └─5401 "nginx: worker process"

Nov 08 20:44:06 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Nov 08 20:44:06 ip-192-168-1-235.ap-south-1.compute.internal nginx[5398]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Nov 08 20:44:06 ip-192-168-1-235.ap-south-1.compute.internal nginx[5398]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Nov 08 20:44:06 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
```

```
[root@ip-192-168-1-235 ~]# ss tnlp | grep :80
Error: an inet prefix is expected rather than "tnlp".
Cannot parse dst/src address.
[root@ip-192-168-1-235 ~]# ss tlnp | grep :80
Error: an inet prefix is expected rather than "tlnp".
Cannot parse dst/src address.
[root@ip-192-168-1-235 ~]# ss -tlnp | grep :80
LISTEN 0      511          0.0.0.0:80        0.0.0.0:*    users:(("nginx",pid=5401,fd=8),("nginx",pid=5400,fd=8))
LISTEN 0      511             [::]:80           [::]:*    users:(("nginx",pid=5401,fd=9),("nginx",pid=5400,fd=9))
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# netstat -tlnp  | grep nginx
tcp        0      0 0.0.0.0:80              0.0.0.0:*               LISTEN      5400/nginx: master
tcp6       0      0 :::80                   :::*                    LISTEN      5400/nginx: master
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
```

```
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

```
[root@ip-192-168-1-235 ~]# curl -I http://localhost
HTTP/1.1 200 OK
Server: nginx/1.30.5
Date: Sun, 08 Nov 2026 20:48:38 GMT
Content-Type: text/html
Content-Length: 896
Last-Modified: Wed, 16 Sep 2026 21:09:29 GMT
Connection: keep-alive
ETag: "6aab0589-380"
Accept-Ranges: bytes
```

**Step 3: Create a custom website**

***Commands:***

```
# Backup default page
cp /usr/share/nginx/html/index.html /usr/share/nginx/html/index.html.bak

# Create custom page
cat > /usr/share/nginx/html/index.html << 'EOF'
<!DOCTYPE html>
<html>
<head>
 <title>Linux Web Server — Capstone Project 2</title>
 <style>
  body { font-family: Arial; background: #1a1a2e; color: #eee; text-align: center; padding: 50px; }
  h1 { color: #16213e; background: #e94560; padding: 20px; border-radius: 8px; }
  .info { background: #16213e; padding: 20px; border-radius: 8px; margin: 20px auto; max-width: 600px; }
  .green { color: #00ff88; }
 </style>
</head>
<body>
 <h1>Linux Web Server</h1>
 <div class="info">
  <p><span class="green">Status:</span> Online and Running</p>
  <p><span class="green">Project:</span> Linux Capstone Project 2</p>
  <p><span class="green">Student:</span> [YourName]</p>
  <p><span class="green">Server:</span> Nginx on Amazon Linux 2023</p>
  <p><span class="green">Purpose:</span> Junior Linux Engineer Capstone</p>
 </div>
</body>
</html>
EOF
```

```
# Create a second page
cat > /usr/share/nginx/html/about.html << 'EOF'
<!DOCTYPE html>
<html>
<body style="font-family:Arial; text-align:center; padding:50px;">
  <h1>About This Server</h1>
  <p>This web server was configured as part of the Linux Capstone Project.</p>
  <p>Technologies used: Nginx | Linux | Firewall | Shell Scripting</p>
  <a href="/">Back to Home</a>
</body>
</html>
EOF

# Reload Nginx to apply changes
systemctl reload nginx

# Test in browser or with curl
curl http://localhost
```

If using EC2, visit http://<public-ip> in your browser — your custom page should load.

***Output:***

```
[root@ip-192-168-1-235 ~]# cp /usr/share/nginx/html/index.html /usr/share/nginx/html/index.html.bak
[root@ip-192-168-1-235 ~]# cat > /usr/share/nginx/html/index.html << 'EOF'
> <!DOCTYPE html>
<html>
<head>
  <title>Linux Web Server — Capstone Project 2</title>
  <style>
    body { font-family: Arial; background: #1a1a2e; color: #eee; text-align: center; padding: 50px; }
    h1 { color: #16213e; background: #e94560; padding: 20px; border-radius: 8px; }
    .info { background: #16213e; padding: 20px; border-radius: 8px; margin: 20px auto; max-width: 600px; }
    .green { color: #00ff88; }
  </style>
</head>
<body>
  <h1>Linux Web Server</h1>
  <div class="info">
    <p><span class="green">Status:</span> Online and Running</p>
    <p><span class="green">Project:</span> Linux Capstone Project 2</p>
    <p><span class="green">Student:</span> [YourName]</p>
    <p><span class="green">Server:</span> Nginx on Amazon Linux 2023</p>
    <p><span class="green">Purpose:</span> Junior Linux Engineer Capstone</p>
  </div>
</body>
</html>
EOF
[root@ip-192-168-1-235 ~]# cat > /usr/share/nginx/html/about.html << 'EOF'
> <!DOCTYPE html>
<html>
<body style="font-family:Arial; text-align:center; padding:50px;">
  <h1>About This Server</h1>
  <p>This web server was configured as part of the Linux Capstone Project.</p>
  <p>Technologies used: Nginx | Linux | Firewall | Shell Scripting</p>
  <a href="/">Back to Home</a>
</body>
</html>
EOF
[root@ip-192-168-1-235 ~]# systemctl reload nginx
[root@ip-192-168-1-235 ~]# curl http://localhost
<!DOCTYPE html>
<html>
<head>
  <title>Linux Web Server — Capstone Project 2</title>
  <style>
    body { font-family: Arial; background: #1a1a2e; color: #eee; text-align: center; padding: 50px; }
    h1 { color: #16213e; background: #e94560; padding: 20px; border-radius: 8px; }
```

```
    h1 { color: #16213e; background: #e94560; padding: 20px; border-radius: 8px; }
    .info { background: #16213e; padding: 20px; border-radius: 8px; margin: 20px auto; max-width: 600px; }
    .green { color: #00ff88; }
  </style>
</head>
<body>
  <h1>Linux Web Server</h1>
  <div class="info">
    <p><span class="green">Status:</span> Online and Running</p>
    <p><span class="green">Project:</span> Linux Capstone Project 2</p>
    <p><span class="green">Student:</span> [YourName]</p>
    <p><span class="green">Server:</span> Nginx on Amazon Linux 2023</p>
    <p><span class="green">Purpose:</span> Junior Linux Engineer Capstone</p>
  </div>
</body>
</html>
[root@ip-192-168-1-235 ~]#
```

**Step 4: Explore Nginx configuration**

***Commands***

# View main Nginx config
cat /etc/nginx/nginx.conf

# View configuration directory
ls -la /etc/nginx/

# Check Nginx config syntax
nginx -t

# View default server block
cat /etc/nginx/conf.d/default.conf 2>/dev/null || cat /etc/nginx/sites-enabled/default 2>/dev/null

# View Nginx logs location
grep "access_log\|error_log" /etc/nginx/nginx.conf

**Output:**

```
[root@ip-192-168-1-235 ~]# cat /etc/nginx/nginx.conf
# For more information on configuration, see:
#   * Official English Documentation: http://nginx.org/en/docs/
#   * Official Russian Documentation: http://nginx.org/ru/docs/

user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

# Load dynamic modules. See /usr/share/doc/nginx/README.dynamic.
include /usr/share/nginx/modules/*.conf;

events {
    worker_connections 1024;
}

http {
    log_format  main  '$remote_addr - $remote_user [$time_local] "$request" '
                      '$status $body_bytes_sent "$http_referer" '
                      '"$http_user_agent" "$http_x_forwarded_for"';

    access_log  /var/log/nginx/access.log  main;

    sendfile            on;
    tcp_nopush          on;
    keepalive_timeout   65;
    types_hash_max_size 4096;

    include             /etc/nginx/mime.types;
    default_type        application/octet-stream;

    # Load modular configuration files from the /etc/nginx/conf.d directory.
    # See http://nginx.org/en/docs/ngx_core_module.html#include
    # for more information.
    include /etc/nginx/conf.d/*.conf;

    server {
        listen       80;
        listen       [::]:80;
        server_name  _;
        root         /usr/share/nginx/html;

        # Load configuration files for the default server block.
        include /etc/nginx/default.d/*.conf;
```

```
#
#        # Load configuration files for the default server block.
#        include /etc/nginx/default.d/*.conf;
#
#        error_page 404 /404.html;
#        location = /404.html {
#        }
#
#        error_page 500 502 503 504 /50x.html;
#        location = /50x.html {
#        }
#    }
}
[root@ip-192-168-1-235 ~]# ls -la /etc/nginx/
total 128
drwxr-xr-x.  4 root root 16384 Nov  8 20:38 .
drwxr-xr-x. 79 root root 16384 Nov  8 20:38 ..
drwxr-xr-x.  2 root root     6 Sep 16 21:09 conf.d
drwxr-xr-x.  2 root root     6 Sep 16 21:09 default.d
-rw-r--r--.  1 root root  1077 Sep 16 21:09 fastcgi.conf
-rw-r--r--.  1 root root  1077 Sep 16 21:09 fastcgi.conf.default
-rw-r--r--.  1 root root  1007 Sep 16 21:09 fastcgi_params
-rw-r--r--.  1 root root  1007 Sep 16 21:09 fastcgi_params.default
-rw-r--r--.  1 root root  2837 Sep 16 21:09 koi-utf
-rw-r--r--.  1 root root  2223 Sep 16 21:09 koi-win
-rw-r--r--.  1 root root 35272 Feb  1  2023 mime.types
-rw-r--r--.  1 root root  5349 Sep 16 21:09 mime.types.default
-rw-r--r--.  1 root root  2317 Sep 16 21:09 nginx.conf
-rw-r--r--.  1 root root  2630 Sep 16 21:09 nginx.conf.default
-rw-r--r--.  1 root root   636 Sep 16 21:09 scgi_params
-rw-r--r--.  1 root root   636 Sep 16 21:09 scgi_params.default
-rw-r--r--.  1 root root   664 Sep 16 21:09 uwsgi_params
-rw-r--r--.  1 root root   664 Sep 16 21:09 uwsgi_params.default
-rw-r--r--.  1 root root  3611 Sep 16 21:09 win-utf
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# cat /etc/nginx/conf.d/default.conf 2>/dev/null || cat /etc/nginx/sites-enabled/default 2>/dev/null

[root@ip-192-168-1-235 ~]# grep "access_log\|error_log" /etc/nginx/nginx.conf
error_log /var/log/nginx/error.log notice;
    access_log  /var/log/nginx/access.log  main;
[root@ip-192-168-1-235 ~]#
```

## Phase 2 — Firewall Configuration

### Step 5: Configure firewall with firewalld

**Commands:**

# Install firewalld if not present
yum install -y firewalld

# Start and enable firewalld
systemctl start firewalld
systemctl enable firewalld
systemctl status firewalld

# Check current firewall rules
firewall-cmd --list-all

# Check default zone
firewall-cmd --get-default-zone

# Allow HTTP (port 80)
firewall-cmd --permanent --add-service=http
firewall-cmd --permanent --add-service=ssh

# Allow a custom port (8080 for testing)
firewall-cmd --permanent --add-port=8080/tcp

# Reload firewall to apply permanent rules
firewall-cmd --reload

# Verify rules applied
firewall-cmd --list-all

# Remove the test port
firewall-cmd --permanent --remove-port=8080/tcp
firewall-cmd --reload
firewall-cmd --list-all

**Output**:

```
[root@ip-192-168-1-235 ~]# yum install firewalld -y
Last metadata expiration check: 23:01:53 ago on Sun Nov  8 20:08:48 2026.
Dependencies resolved.
================================================================================================================================================
 Package                                 Architecture              Version                                 Repository                        Size
================================================================================================================================================
Installing:
 firewalld                               noarch                    1.2.3-1.amzn2023.0.2                    amazonlinux                      450 k
Installing dependencies:
 firewalld-filesystem                    noarch                    1.2.3-1.amzn2023.0.2                    amazonlinux                      8.8 k
 gobject-introspection                   x86_64                    1.82.0-1.amzn2023                       amazonlinux                      122 k
 ipset                                   x86_64                    7.11-1.amzn2023.0.3                     amazonlinux                       40 k
 ipset-libs                              x86_64                    7.11-1.amzn2023.0.3                     amazonlinux                       67 k
 iptables-libs                           x86_64                    1.8.8-3.amzn2023.0.2                    amazonlinux                      401 k
 iptables-nft                            x86_64                    1.8.8-3.amzn2023.0.2                    amazonlinux                      183 k
 libnetfilter_conntrack                  x86_64                    1.0.8-2.amzn2023.0.2                    amazonlinux                       58 k
 libnfnetlink                            x86_64                    1.0.1-19.amzn2023.0.2                   amazonlinux                       30 k
 libnftnl                                x86_64                    1.2.2-2.amzn2023.0.2                    amazonlinux                       84 k
 nftables                                x86_64                    1:1.0.4-3.amzn2023.0.3                  amazonlinux                      397 k
 python3-firewall                        noarch                    1.2.3-1.amzn2023.0.2                    amazonlinux                      355 k
 python3-gobject-base                    x86_64                    3.48.2-3.amzn2023.0.2                   amazonlinux                      323 k
 python3-nftables                        x86_64                    1:1.0.4-3.amzn2023.0.3                  amazonlinux                       16 k
Installing weak dependencies:
 libcap-ng-python3                       x86_64                    0.8.2-4.amzn2023.0.2                    amazonlinux                       30 k

Transaction Summary
================================================================================================================================================
Install  15 Packages

Total download size: 2.5 M
Installed size: 11 M
Downloading Packages:
(1/15): gobject-introspection-1.82.0-1.amzn2023.x86_64.rpm                                                4.2 MB/s | 122 kB     00:00
(2/15): ipset-7.11-1.amzn2023.0.3.x86_64.rpm                                                              1.6 MB/s |  40 kB     00:00
(3/15): firewalld-filesystem-1.2.3-1.amzn2023.0.2.noarch.rpm                                              163 kB/s | 8.8 kB     00:00
(4/15): ipset-libs-7.11-1.amzn2023.0.3.x86_64.rpm                                                         2.2 MB/s |  67 kB     00:00
(5/15): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm                                                      11 MB/s | 401 kB     00:00
(6/15): firewalld-1.2.3-1.amzn2023.0.2.noarch.rpm                                                         4.2 MB/s | 450 kB     00:00
(7/15): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm                                                      5.7 MB/s | 183 kB     00:00
(8/15): libcap-ng-python3-0.8.2-4.amzn2023.0.2.x86_64.rpm                                                 733 kB/s |  30 kB     00:00
(9/15): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm                                            1.8 MB/s |  58 kB     00:00
(10/15): libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64.rpm                                                    979 kB/s |  30 kB     00:00
(11/15): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm                                                         3.1 MB/s |  84 kB     00:00
(12/15): nftables-1.0.4-3.amzn2023.0.3.x86_64.rpm                                                          11 MB/s | 397 kB     00:00
(13/15): python3-nftables-1.0.4-3.amzn2023.0.3.x86_64.rpm                                                 323 kB/s |  16 kB     00:00
```

```
Complete!
[root@ip-192-168-1-235 ~]# systemctl start firewalld
[root@ip-192-168-1-235 ~]# systemctl enable firewalld
[root@ip-192-168-1-235 ~]# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
     Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-11-09 19:11:01 UTC; 14s ago
       Docs: man:firewalld(1)
   Main PID: 46481 (firewalld)
      Tasks: 2 (limit: 1113)
     Memory: 25.5M
        CPU: 331ms
     CGroup: /system.slice/firewalld.service
             └─46481 /usr/bin/python3 -s /usr/sbin/firewalld --nofork --nopid

Nov 09 19:11:01 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Starting firewalld.service - firewalld - dynamic firewall daemon...
Nov 09 19:11:01 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Started firewalld.service - firewalld - dynamic firewall daemon.
[root@ip-192-168-1-235 ~]# firewall-cmd --list-all
public
  target: default
  icmp-block-inversion: no
  interfaces:
  sources:
  services: dhcpv6-client mdns ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

```
[root@ip-192-168-1-235 ~]# firewall-cmd --get-default-zone
public
[root@ip-192-168-1-235 ~]# firewall-cmd --permanent --add-service=http
success
[root@ip-192-168-1-235 ~]# firewall-cmd --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[root@ip-192-168-1-235 ~]# firewall-cmd --permanent --add-port=8080/tcp
success
[root@ip-192-168-1-235 ~]# firewall-cmd --reload
success
[root@ip-192-168-1-235 ~]# firewall-cmd --list-all
public
  target: default
  icmp-block-inversion: no
  interfaces:
  sources:
  services: dhcpv6-client http mdns ssh
  ports: 8080/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@ip-192-168-1-235 ~]# firewall-cmd --permanent --remove-port=8080/tcp
success
[root@ip-192-168-1-235 ~]# firewall-cmd --reload
success
[root@ip-192-168-1-235 ~]# firewall-cmd --list-all
public
  target: default
  icmp-block-inversion: no
  interfaces:
  sources:
  services: dhcpv6-client http mdns ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

**Step 6: Test firewall blocking**

***Commnads:***

# Try to access a blocked port from the server itself
curl http://localhost:8080  # should fail/timeout after removal

# Check what ports are open
ss -tlnp
firewall-cmd --list-ports
firewall-cmd --list-services

# Block a specific IP (example — use a test IP)
firewall-cmd --permanent --add-rich-rule='rule family="ipv4" source address="192.168.100.100" reject'
firewall-cmd --reload
firewall-cmd --list-rich-rules

# Remove the IP block
firewall-cmd --permanent --remove-rich-rule='rule family="ipv4" source address="192.168.100.100" reject'
firewall-cmd –reload

***Output:***

```
[root@ip-192-168-1-235 ~]# ss -tlnp
State       Recv-Q      Send-Q           Local Address:Port           Peer Address:Port      Process
LISTEN      0           128                    0.0.0.0:22                  0.0.0.0:*          users:(("sshd",pid=2617,fd=7))
LISTEN      0           511                    0.0.0.0:80                  0.0.0.0:*          users:(("nginx",pid=5980,fd=8),("nginx",pid=5400,fd=8))
LISTEN      0           128                       [::]:22                     [::]:*          users:(("sshd",pid=2617,fd=8))
LISTEN      0           511                       [::]:80                     [::]:*          users:(("nginx",pid=5980,fd=9),("nginx",pid=5400,fd=9))
```

```
[root@ip-192-168-1-235 ~]# firewall-cmd --list-ports

[root@ip-192-168-1-235 ~]# firewall-cmd --list-services
dhcpv6-client http mdns ssh
[root@ip-192-168-1-235 ~]# firewall-cmd --permanent --add-rich-rule='rule family="ipv4" source address="192.168.100.100" reject'
success
[root@ip-192-168-1-235 ~]# firewall-cmd --reload
success
[root@ip-192-168-1-235 ~]# firewall-cmd --list-rich-rules
rule family="ipv4" source address="192.168.100.100" reject
[root@ip-192-168-1-235 ~]# firewall-cmd --permanent --remove-rich-rule='rule family="ipv4" source address="192.168.100.100" reject'
success
[root@ip-192-168-1-235 ~]# firewall-cmd --reload
success
[root@ip-192-168-1-235 ~]#
```

**Phase 3 — Log Monitoring & Network Commands**

**Step 7: Generate and monitor Nginx logs**

***Commands:***

```
# Generate web traffic to create log entries
curl http://localhost/
curl http://localhost/about.html
curl http://localhost/nonexistent.html    # generates 404
curl http://localhost/
curl -A "TestBrowser/1.0" http://localhost/

# View access logs
tail -20 /var/log/nginx/access.log

# View error logs
tail -20 /var/log/nginx/error.log

# Real-time log monitoring (run then make requests in another terminal)
tail -f /var/log/nginx/access.log &

# Generate more traffic
for i in {1..5}; do curl -s http://localhost/ > /dev/null; done

# Stop the tail
kill %1

# Analyze logs with grep and awk
grep "404" /var/log/nginx/access.log
grep "200" /var/log/nginx/access.log | wc -l
awk '{print $1}' /var/log/nginx/access.log | sort | uniq -c | sort -rn
awk '{print $9}' /var/log/nginx/access.log | sort | uniq -c
```

***Output:***

```
[root@ip-192-168-1-235 ~]# curl http://localhost/
<!DOCTYPE html>
<html>
<head>
  <title>Linux Web Server - Capstone Project 2</title>
  <style>
    body { font-family: Arial; background: #1a1a2e; color: #eee; text-align: center; padding: 50px; }
    h1 { color: #16213e; background: #e94560; padding: 20px; border-radius: 8px; }
    .info { background: #16213e; padding: 20px; border-radius: 8px; margin: 20px auto; max-width: 600px; }
    .green { color: #00ff88; }
  </style>
</head>
<body>
  <h1>Linux Web Server</h1>
  <div class="info">
    <p><span class="green">Status:</span> Online and Running</p>
    <p><span class="green">Project:</span> Linux Capstone Project 2</p>
    <p><span class="green">Student:</span> [YourName]</p>
    <p><span class="green">Server:</span> Nginx on Amazon Linux 2023</p>
    <p><span class="green">Purpose:</span> Junior Linux Engineer Capstone</p>
  </div>
</body>
</html>
[root@ip-192-168-1-235 ~]# curl http://localhost/about.html
<!DOCTYPE html>
<html>
<body style="font-family:Arial; text-align:center; padding:50px;">
  <h1>About This Server</h1>
  <p>This web server was configured as part of the Linux Capstone Project.</p>
  <p>Technologies used: Nginx | Linux | Firewall | Shell Scripting</p>
  <a href="/">Back to Home</a>
</body>
</html>
[root@ip-192-168-1-235 ~]# curl http://localhost/nonexistent.html
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN" "http://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd">

<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en">
    <head>
        <title>The page is not found</title>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
        <style type="text/css">
            /*<![CDATA[*/
            body {
                background-color: #fff;
                color: #000;
                color: #c00;
            }
            :visited {
                color: #c00;
            }
            a:hover {
                color: #f50;
            }
            h1 {
                text-align: center;
                margin: 0;
                padding: 0.6em 2em 0.4em;
                background-color: #294172;
                color: #fff;
                font-weight: normal;
                font-size: 1.75em;
                border-bottom: 2px solid #000;
            }
            h1 strong {
                font-weight: bold;
                font-size: 1.5em;
            }
            h2 {
                text-align: center;
                background-color: #3C6EB4;
                font-size: 1.1em;
                font-weight: bold;
                color: #fff;
                margin: 0;
                padding: 0.5em;
                border-bottom: 2px solid #294172;
            }
            h3 {
                text-align: center;
                background-color: #ff0000;
                padding: 0.5em;
                color: #fff;
            }
            hr {
                display: none;
            }
            .content {
                padding: 1em 5em;
            }
            .alert {
                padding: 1em 5em;
            }
            .alert {
                border: 2px solid #000;
            }

            img {
                border: 2px solid #fff;
                padding: 2px;
                margin: 2px;
            }
            a:hover img {
                border: 2px solid #294172;
            }
            .logos {
                margin: 1em;
                text-align: center;
            }
            /*]]>*/
        </style>
    </head>

    <body>
        <h1><strong>nginx error!</strong></h1>

        <div class="content">

            <h3>The page you are looking for is not found.</h3>

            <div class="alert">
                <h2>Website Administrator</h2>
                <div class="content">
                    <p>Something has triggered missing webpage on your
                    website. This is the default 404 error page for
                    <strong>nginx</strong> that is distributed with
                    Amazon Linux. It is located
                    <tt>/usr/share/nginx/html/404.html</tt></p>

                    <p>You should customize this error page for your own
                    site or edit the <tt>error_page</tt> directive in
                    the <strong>nginx</strong> configuration file
                    <tt>/etc/nginx/nginx.conf</tt>.</p>
                </div>
                <div class="logos">
                  <a href="http://nginx.net/"><img
                    src="/nginx-logo.png"
                    alt="[ Powered by nginx ]"
                    width="121" height="32" /></a>
                </div>
            </div>
        </div>
    </body>
</html>
```

```
[root@ip-192-168-1-235 ~]# curl http://localhost/
<!DOCTYPE html>
<html>
<head>
  <title>Linux Web Server - Capstone Project 2</title>
  <style>
    body { font-family: Arial; background: #1a1a2e; color: #eee; text-align: center; padding: 50px; }
    h1 { color: #16213e; background: #e94560; padding: 20px; border-radius: 8px; }
    .info { background: #16213e; padding: 20px; border-radius: 8px; margin: 20px auto; max-width: 600px; }
    .green { color: #00ff88; }
  </style>
</head>
<body>
  <h1>Linux Web Server</h1>
  <div class="info">
    <p><span class="green">Status:</span> Online and Running</p>
    <p><span class="green">Project:</span> Linux Capstone Project 2</p>
    <p><span class="green">Student:</span> [YourName]</p>
    <p><span class="green">Server:</span> Nginx on Amazon Linux 2023</p>
    <p><span class="green">Purpose:</span> Junior Linux Engineer Capstone</p>
  </div>
</body>
</html>
[root@ip-192-168-1-235 ~]# curl -A "TestBrowser/1.0" http://localhost/
<!DOCTYPE html>
<html>
<head>
  <title>Linux Web Server - Capstone Project 2</title>
  <style>
    body { font-family: Arial; background: #1a1a2e; color: #eee; text-align: center; padding: 50px; }
    h1 { color: #16213e; background: #e94560; padding: 20px; border-radius: 8px; }
    .info { background: #16213e; padding: 20px; border-radius: 8px; margin: 20px auto; max-width: 600px; }
    .green { color: #00ff88; }
  </style>
</head>
<body>
  <h1>Linux Web Server</h1>
  <div class="info">
    <p><span class="green">Status:</span> Online and Running</p>
    <p><span class="green">Project:</span> Linux Capstone Project 2</p>
    <p><span class="green">Student:</span> [YourName]</p>
    <p><span class="green">Server:</span> Nginx on Amazon Linux 2023</p>
    <p><span class="green">Purpose:</span> Junior Linux Engineer Capstone</p>
  </div>
</body>
</html>
[root@ip-192-168-1-235 ~]# tail -20 /var/log/nginx/access.log
::1 - - [08/Nov/2026:20:47:54 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.21.0" "-"
::1 - - [08/Nov/2026:20:48:21 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.21.0" "-"
::1 - - [08/Nov/2026:20:48:38 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.21.0" "-"
::1 - - [08/Nov/2026:20:58:31 +0000] "GET / HTTP/1.1" 200 866 "-" "curl/8.21.0" "-"
::1 - - [09/Nov/2026:19:28:26 +0000] "GET / HTTP/1.1" 200 866 "-" "curl/8.21.0" "-"
::1 - - [09/Nov/2026:19:29:10 +0000] "GET /about.html HTTP/1.1" 200 316 "-" "curl/8.21.0" "-"
::1 - - [09/Nov/2026:19:29:45 +0000] "GET /nonexistent.html HTTP/1.1" 404 3464 "-" "curl/8.21.0" "-"
::1 - - [09/Nov/2026:19:30:03 +0000] "GET / HTTP/1.1" 200 866 "-" "curl/8.21.0" "-"
::1 - - [09/Nov/2026:19:30:16 +0000] "GET / HTTP/1.1" 200 866 "-" "TestBrowser/1.0" "-"
[root@ip-192-168-1-235 ~]# tail -20 /var/log/nginx/error.log
2026/11/08 20:44:06 [notice] 5399#5399: using the "epoll" event method
2026/11/08 20:44:06 [notice] 5399#5399: nginx/1.30.5
2026/11/08 20:44:06 [notice] 5399#5399: built by gcc 11.5.0 20240719 (Red Hat 11.5.0-5) (GCC)
2026/11/08 20:44:06 [notice] 5399#5399: OS: Linux 6.18.51-120.163.amzn2023.x86_64
2026/11/08 20:44:06 [notice] 5399#5399: getrlimit(RLIMIT_NOFILE): 65535:65535
2026/11/08 20:44:06 [notice] 5400#5400: start worker processes
2026/11/08 20:44:06 [notice] 5400#5400: start worker process 5401
2026/11/08 20:58:23 [notice] 5979#5979: signal process started
2026/11/08 20:58:23 [notice] 5400#5400: signal 1 (SIGHUP) received from 5979, reconfiguring
2026/11/08 20:58:23 [notice] 5400#5400: reconfiguring
2026/11/08 20:58:23 [notice] 5400#5400: using the "epoll" event method
2026/11/08 20:58:23 [notice] 5400#5400: start worker processes
2026/11/08 20:58:23 [notice] 5400#5400: start worker process 5980
2026/11/08 20:58:23 [notice] 5401#5401: gracefully shutting down
2026/11/08 20:58:23 [notice] 5401#5401: exiting
2026/11/08 20:58:23 [notice] 5401#5401: exit
2026/11/08 20:58:23 [notice] 5400#5400: signal 17 (SIGCHLD) received from 5401
2026/11/08 20:58:23 [notice] 5400#5400: worker process 5401 exited with code 0
2026/11/08 20:58:23 [notice] 5400#5400: signal 29 (SIGIO) received
2026/11/09 19:29:45 [error] 5980#5980: *7 open() "/usr/share/nginx/html/nonexistent.html" failed (2: No such file or directory), client: ::1, server: _, req
uest: "GET /nonexistent.html HTTP/1.1", host: "localhost"
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# tail -f /var/log/nginx/access.log &
[1] 47420
::1 - - [08/Nov/2026:20:47:54 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.21.0" "-"
[root@ip-192-168-1-235 ~]# ::1 - - [08/Nov/2026:20:48:21 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.21.0" "-"
::1 - - [08/Nov/2026:20:48:38 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.21.0" "-"
::1 - - [08/Nov/2026:20:58:31 +0000] "GET / HTTP/1.1" 200 866 "-" "curl/8.21.0" "-"
::1 - - [09/Nov/2026:19:28:26 +0000] "GET / HTTP/1.1" 200 866 "-" "curl/8.21.0" "-"
::1 - - [09/Nov/2026:19:29:10 +0000] "GET /about.html HTTP/1.1" 200 316 "-" "curl/8.21.0" "-"
::1 - - [09/Nov/2026:19:29:45 +0000] "GET /nonexistent.html HTTP/1.1" 404 3464 "-" "curl/8.21.0" "-"
::1 - - [09/Nov/2026:19:30:03 +0000] "GET / HTTP/1.1" 200 866 "-" "curl/8.21.0" "-"
::1 - - [09/Nov/2026:19:30:16 +0000] "GET / HTTP/1.1" 200 866 "-" "TestBrowser/1.0" "-"
[root@ip-192-168-1-235 ~]# for i in {1..5}; do curl -s http://localhost/ > /dev/null; done
::1 - - [09/Nov/2026:19:32:07 +0000] "GET / HTTP/1.1" 200 866 "-" "curl/8.21.0" "-"
::1 - - [09/Nov/2026:19:32:07 +0000] "GET / HTTP/1.1" 200 866 "-" "curl/8.21.0" "-"
::1 - - [09/Nov/2026:19:32:07 +0000] "GET / HTTP/1.1" 200 866 "-" "curl/8.21.0" "-"
::1 - - [09/Nov/2026:19:32:07 +0000] "GET / HTTP/1.1" 200 866 "-" "curl/8.21.0" "-"
::1 - - [09/Nov/2026:19:32:07 +0000] "GET / HTTP/1.1" 200 866 "-" "curl/8.21.0" "-"
[root@ip-192-168-1-235 ~]# kill %1
[root@ip-192-168-1-235 ~]# grep "404" /var/log/nginx/access.log
::1 - - [09/Nov/2026:19:29:45 +0000] "GET /nonexistent.html HTTP/1.1" 404 3464 "-" "curl/8.21.0" "-"
[1]+  Terminated              tail -f /var/log/nginx/access.log
[root@ip-192-168-1-235 ~]# grep "200" /var/log/nginx/access.log | wc -l
13
[root@ip-192-168-1-235 ~]# awk '{print $1}' /var/log/nginx/access.log | sort | uniq -c | sort -rn
     14 ::1
[root@ip-192-168-1-235 ~]# awk '{print $9}' /var/log/nginx/access.log | sort | uniq -c
     13 200
      1 404
[root@ip-192-168-1-235 ~]#
```

**Step 8: System log monitoring**

***Commands:***

```
# View system logs
tail -50 /var/log/messages       # CentOS/Amazon Linux
# tail -50 /var/log/syslog      # Ubuntu

# View authentication logs
tail -20 /var/log/secure         # CentOS/Amazon Linux
# tail -20 /var/log/auth.log     # Ubuntu

# Filter for specific events
grep "nginx" /var/log/messages
grep "Failed" /var/log/secure 2>/dev/null || grep "Failed" /var/log/auth.log 2>/dev/null

# Monitor logs live
journalctl -u nginx -f &
sleep 3
curl http://localhost
kill %1

# View journal for specific service
journalctl -u nginx --since "1 hour ago"
journalctl -u firewalld --since "today"
```

***Output:***

```
[root@ip-192-168-1-235 ~]# tail -50 /var/log/messages
tail: cannot open '/var/log/messages' for reading: No such file or directory
[root@ip-192-168-1-235 ~]# tail -20 /var/log/secure
tail: cannot open '/var/log/secure' for reading: No such file or directory
[root@ip-192-168-1-235 ~]# tail -20 /var/log/auth.log
tail: cannot open '/var/log/auth.log' for reading: No such file or directory
[root@ip-192-168-1-235 ~]# grep "nginx" /var/log/messages
grep: /var/log/messages: No such file or directory
[root@ip-192-168-1-235 ~]# grep "Failed" /var/log/secure 2>/dev/null || grep "Failed" /var/log/auth.log 2>/dev/null
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# journalctl -u nginx -f &
[1] 48164
Nov 08 20:41:16 ip-192-168-1-235.ap-south-1.compute.internal nginx[5197]: nginx: [emerg] still could not bind()
[root@ip-192-168-1-235 ~]# Nov 08 20:41:16 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: nginx.service: Control process exited, code=exited, stat
us=1/FAILURE
Nov 08 20:41:16 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: nginx.service: Failed with result 'exit-code'.
Nov 08 20:41:16 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Failed to start nginx.service - The nginx HTTP and reverse proxy server.
Nov 08 20:44:06 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Nov 08 20:44:06 ip-192-168-1-235.ap-south-1.compute.internal nginx[5398]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Nov 08 20:44:06 ip-192-168-1-235.ap-south-1.compute.internal nginx[5398]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Nov 08 20:44:06 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
Nov 08 20:58:23 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Reloading nginx.service - The nginx HTTP and reverse proxy server...
Nov 08 20:58:23 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Reloaded nginx.service - The nginx HTTP and reverse proxy server.

[root@ip-192-168-1-235 ~]# sleep 3
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# curl http://localhost
<!DOCTYPE html>
<html>
<head>
  <title>Linux Web Server - Capstone Project 2</title>
  <style>
    body { font-family: Arial; background: #1a1a2e; color: #eee; text-align: center; padding: 50px; }
    h1 { color: #16213e; background: #e94560; padding: 20px; border-radius: 8px; }
    .info { background: #16213e; padding: 20px; border-radius: 8px; margin: 20px auto; max-width: 600px; }
    .green { color: #00ff88; }
  </style>
</head>
<body>
  <h1>Linux Web Server</h1>
  <div class="info">
    <p><span class="green">Status:</span> Online and Running</p>
    <p><span class="green">Project:</span> Linux Capstone Project 2</p>
    <p><span class="green">Student:</span> [YourName]</p>
    <p><span class="green">Server:</span> Nginx on Amazon Linux 2023</p>
```

```
    <p><span class="green">Student:</span> [YourName]</p>
    <p><span class="green">Server:</span> Nginx on Amazon Linux 2023</p>
    <p><span class="green">Purpose:</span> Junior Linux Engineer Capstone</p>
  </div>
</body>
</html>
[root@ip-192-168-1-235 ~]# kill %1
[root@ip-192-168-1-235 ~]# journalctl -u nginx --since "1 hour ago"
-- No entries --
[1]+  Terminated              journalctl -u nginx -f
[root@ip-192-168-1-235 ~]# journalctl -u firewalld --since "today"
Nov 09 19:11:01 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Starting firewalld.service - firewalld - dynamic firewall daemon...
Nov 09 19:11:01 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Started firewalld.service - firewalld - dynamic firewall daemon.
Nov 09 19:12:57 ip-192-168-1-235.ap-south-1.compute.internal firewalld[46481]: WARNING: ALREADY_ENABLED: ssh
[root@ip-192-168-1-235 ~]#
```

**Step 9: Network commands**

***Commands:***

# Check network interfaces
ip addr show
ip link show
ifconfig 2>/dev/null || ip addr

# Check routing table
ip route show
route -n 2>/dev/null || ip route

# Test connectivity
ping -c 4 google.com
ping -c 4 8.8.8.8

# DNS lookup
nslookup google.com
dig google.com
dig google.com +short

# Download a file with wget and curl
wget -O /tmp/test_download.html http://localhost/
curl -o /tmp/curl_download.html http://localhost/
ls -lh /tmp/test_download.html /tmp/curl_download.html

# Check open ports
ss -tlnp
ss -tlnp | grep nginx
ss -tlnp | grep :80

# Check network stats
netstat -s 2>/dev/null || ss -s

# Trace route
traceroute google.com 2>/dev/null || tracepath google.com

## *Output:*

```
[root@ip-192-168-1-235 ~]# ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enX0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc fq_codel state UP group default qlen 1000
    link/ether 0a:ff:e8:d6:2e:8d brd ff:ff:ff:ff:ff:ff
    altname eni-0ed654e8786b155b4
    altname device-number-0.0
    inet 192.168.1.235/25 metric 512 brd 192.168.1.255 scope global dynamic enX0
       valid_lft 2637sec preferred_lft 2637sec
    inet6 fe80::8ff:e8ff:fed6:2e8d/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enX0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 0a:ff:e8:d6:2e:8d brd ff:ff:ff:ff:ff:ff
    altname eni-0ed654e8786b155b4
    altname device-number-0.0
[root@ip-192-168-1-235 ~]# ifconfig 2>/dev/null || ip addr
enX0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 9001
        inet 192.168.1.235  netmask 255.255.255.128  broadcast 192.168.1.255
        inet6 fe80::8ff:e8ff:fed6:2e8d  prefixlen 64  scopeid 0x20<link>
        ether 0a:ff:e8:d6:2e:8d  txqueuelen 1000  (Ethernet)
        RX packets 81590  bytes 28753944 (27.4 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 74305  bytes 30234792 (28.8 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 178  bytes 32188 (31.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 178  bytes 32188 (31.4 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[root@ip-192-168-1-235 ~]# ip route show
default via 192.168.1.129 dev enX0 proto dhcp src 192.168.1.235 metric 512
192.168.1.2 via 192.168.1.129 dev enX0 proto dhcp src 192.168.1.235 metric 512
192.168.1.129 dev enX0 proto dhcp scope link src 192.168.1.235 metric 512
[root@ip-192-168-1-235 ~]# route -n 2>/dev/null || ip route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         192.168.1.129   0.0.0.0         UG    512    0        0 enX0
192.168.1.2     192.168.1.129   255.255.255.255 UGH   512    0        0 enX0
192.168.1.128   0.0.0.0         255.255.255.128 U     512    0        0 enX0
192.168.1.129   0.0.0.0         255.255.255.255 UH    512    0        0 enX0
[root@ip-192-168-1-235 ~]# ping -c 4 google.com
PING google.com (192.178.173.113) 56(84) bytes of data.
64 bytes from lcbome-in-f113.1e100.net (192.178.173.113): icmp_seq=1 ttl=113 time=1.44 ms
64 bytes from lcbome-in-f113.1e100.net (192.178.173.113): icmp_seq=2 ttl=113 time=1.26 ms
64 bytes from lcbome-in-f113.1e100.net (192.178.173.113): icmp_seq=3 ttl=113 time=1.25 ms
64 bytes from lcbome-in-f113.1e100.net (192.178.173.113): icmp_seq=4 ttl=113 time=1.27 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.248/1.302/1.435/0.077 ms
[root@ip-192-168-1-235 ~]# ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=1.06 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=1.07 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=117 time=1.06 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=117 time=1.08 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 1.061/1.069/1.083/0.008 ms
[root@ip-192-168-1-235 ~]# nslookup google.com
Server:         192.168.1.2
Address:        192.168.1.2#53

Non-authoritative answer:
Name:   google.com
Address: 192.178.173.102
Name:   google.com
Address: 192.178.173.139
Name:   google.com
Address: 192.178.173.100
Name:   google.com
Address: 192.178.173.138
Name:   google.com
Address: 192.178.173.101
Name:   google.com
Name:   google.com
Address: 192.178.173.101
Name:   google.com
Address: 192.178.173.113
Name:   google.com
Address: 2404:6800:4000:100e::64
Name:   google.com
Address: 2404:6800:4000:100e::65
Name:   google.com
Address: 2404:6800:4000:100e::66
Name:   google.com
Address: 2404:6800:4000:100e::8a

[root@ip-192-168-1-235 ~]# dig google.com

; <<>> DiG 9.18.50 <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 22466
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1
```

```
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             74      IN      A       192.178.173.113
google.com.             74      IN      A       192.178.173.102
google.com.             74      IN      A       192.178.173.139
google.com.             74      IN      A       192.178.173.100
google.com.             74      IN      A       192.178.173.138
google.com.             74      IN      A       192.178.173.101

;; Query time: 0 msec
;; SERVER: 192.168.1.2#53(192.168.1.2) (UDP)
;; WHEN: Mon Nov 09 19:57:36 UTC 2026
;; MSG SIZE  rcvd: 135

[root@ip-192-168-1-235 ~]# dig google.com +short
192.178.173.101
192.178.173.113
192.178.173.102
192.178.173.139
192.178.173.100
192.178.173.138
[root@ip-192-168-1-235 ~]# wget -O /tmp/test_download.html http://localhost/
--2026-11-09 19:58:25--  http://localhost/
Resolving localhost (localhost)... 127.0.0.1
Connecting to localhost (localhost)|127.0.0.1|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 866 [text/html]
Saving to: ‘/tmp/test_download.html’

/tmp/test_download.html               100%[=========================================================================>]     866  --.-KB/s    in 0s

2026-11-09 19:58:25 (254 MB/s) - ‘/tmp/test_download.html’ saved [866/866]

[root@ip-192-168-1-235 ~]# curl -o /tmp/test_download.html http://localhost/
  % Total    % Received % Xferd  Average Speed  Time    Time    Time   Current
                                 Dload  Upload  Total   Spent   Left   Speed
100    866 100    866   0      0 615.5k      0                              0
[root@ip-192-168-1-235 ~]# curl -o /tmp/curl_download.html http://localhost/
  % Total    % Received % Xferd  Average Speed  Time    Time    Time   Current
                                 Dload  Upload  Total   Spent   Left   Speed
100    866 100    866   0      0 136.0k      0                              0
```

```
[root@ip-192-168-1-235 ~]# ls -lh /tmp/test_download.html /tmp/curl_download.html
-rw-r--r--. 1 root root 866 Nov  9 19:59 /tmp/curl_download.html
-rw-r--r--. 1 root root 866 Nov  9 19:58 /tmp/test_download.html
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# ss -tlnp
State       Recv-Q      Send-Q           Local Address:Port            Peer Address:Port      Process
LISTEN      0           128                    0.0.0.0:22                   0.0.0.0:*          users:(("sshd",pid=2617,fd=7))
LISTEN      0           511                    0.0.0.0:80                   0.0.0.0:*          users:(("nginx",pid=5980,fd=8),("nginx",pid=5400,fd=8))
LISTEN      0           128                       [::]:22                      [::]:*          users:(("sshd",pid=2617,fd=8))
LISTEN      0           511                       [::]:80                      [::]:*          users:(("nginx",pid=5980,fd=9),("nginx",pid=5400,fd=9))
[root@ip-192-168-1-235 ~]# ss -tlnp | grep nginx
LISTEN 0      511          0.0.0.0:80        0.0.0.0:*    users:(("nginx",pid=5980,fd=8),("nginx",pid=5400,fd=8))
LISTEN 0      511             [::]:80           [::]:*    users:(("nginx",pid=5980,fd=9),("nginx",pid=5400,fd=9))
[root@ip-192-168-1-235 ~]# ss -tlnp | grep :80
LISTEN 0      511          0.0.0.0:80        0.0.0.0:*    users:(("nginx",pid=5980,fd=8),("nginx",pid=5400,fd=8))
LISTEN 0      511             [::]:80           [::]:*    users:(("nginx",pid=5980,fd=9),("nginx",pid=5400,fd=9))
[root@ip-192-168-1-235 ~]# netstat -s 2>/dev/null || ss -s
Ip:
    Forwarding: 2
    76483 total packets received
    4 with invalid addresses
    0 forwarded
    0 incoming packets discarded
    76479 incoming packets delivered
    76288 requests sent out
    OutTransmits: 76288
Icmp:
    14 ICMP messages received
    0 input ICMP message failed
    ICMP input histogram:
        destination unreachable: 6
        echo replies: 8
    14 ICMP messages sent
    0 ICMP messages failed
    ICMP output histogram:
        destination unreachable: 6
        echo requests: 8
IcmpMsg:
        InType0: 8
        InType3: 6
        OutType3: 6
        OutType8: 8
Tcp:
    4663 active connection openings
    109 passive connection openings
```

```
Tcp:
    4663 active connection openings
    109 passive connection openings
    27 failed connection attempts
    15 connection resets received
    1 connections established
    70618 segments received
    85732 segments sent out
    1525 segments retransmitted
    0 bad segments received
    83 resets sent
Udp:
    6013 packets received
    6 packets to unknown port received
    0 packet receive errors
    6020 packets sent
    0 receive buffer errors
    0 send buffer errors
```

```
UdpLite:
TcpExt:
    25 resets received for embryonic SYN_RECV sockets
    4 packets pruned from receive queue because of socket buffer overrun
    22 TCP sockets finished time wait in fast timer
    79 delayed acks sent
    Quick ack mode was activated 205 times
    576 packet headers predicted
    10264 acknowledgments not containing data payload received
    5127 predicted acknowledgments
    TCPSackRecovery: 40
    Detected reordering 293 times using SACK
    TCPDSACKUndo: 22
    TCPLostRetransmit: 4
    40 fast retransmits
    TCPTimeouts: 1423
    TCPLossProbes: 96
    TCPBacklogCoalesce: 40
    TCPDSACKOldSent: 272
    TCPDSACKRecv: 88
    TCPDSACKOfoRecv: 4
    21 connections reset due to unexpected data
    6 connections reset due to early user close
    2 connections aborted due to timeout
    TCPDSACKIgnoredNoUndo: 58
    TCPSackShifted: 452
    TCPSackMerged: 67
    TCPSackShiftFallback: 533
    TCPRcvCoalesce: 1458
    TCPOFOQueue: 473
    TCPAutoCorking: 290
    TCPFromZeroWindowAdv: 7
    TCPToZeroWindowAdv: 7
    TCPWantZeroWindowAdv: 1
    TCPSynRetrans: 1385
    TCPOrigDataSent: 71406
    TCPHystartDelayDetect: 2
    TCPHystartDelayCwnd: 246
    TCPACKSkippedSeq: 6
    TCPKeepAlive: 70
    TCPDelivered: 76010
    TCPAckCompressed: 162
    TcpTimeoutRehash: 7
    TCPDSACKRecvSegs: 92
```

```
IpExt:
    InOctets: 28797926
    OutOctets: 29433964
    InNoECTPkts: 84881
    InECT1Pkts: 3
    InECT0Pkts: 5
    InCEPkts: 1
MPTcpExt:
[root@ip-192-168-1-235 ~]# traceroute google.com 2>/dev/null || tracepath google.com
traceroute to google.com (192.178.173.101), 30 hops max, 60 byte packets
 1  242.6.252.5 (242.6.252.5)  1.136 ms 242.6.253.3 (242.6.253.3)  1.433 ms 242.6.253.1 (242.6.253.1)  1.069 ms
 2  240.2.196.1 (240.2.196.1)  1.710 ms  2.256 ms 240.2.196.0 (240.2.196.0)  1.183 ms
 3  99.82.178.53 (99.82.178.53)  0.957 ms 99.82.180.91 (99.82.180.91)  0.938 ms 99.82.178.53 (99.82.178.53)  0.895 ms
 4  lcbome-in-f101.1e100.net (192.178.173.101)  1.960 ms  1.857 ms  1.629 ms
[root@ip-192-168-1-235 ~]#
```

**Phase 4 — Shell Scripting & Automation**

**Step 10: Write a server health check script**

***Commads*:**

cat > /usr/local/bin/server_health_check.sh << 'SCRIPT'
#!/bin/bash

# ==========================================
# Server Health Check Script
# Linux Capstone Project 2
# ==========================================

LOGFILE="/var/log/health_check.log"
TIMESTAMP=$(date '+%Y-%m-%d %H:%M:%S')
PASS=0
FAIL=0

log() {

```
    echo "[$TIMESTAMP] $1" | tee -a $LOGFILE
}

check_service() {
    SERVICE=$1
    if systemctl is-active --quiet $SERVICE; then
        log " PASS: $SERVICE is running"
        ((PASS++))
    else
        log " FAIL: $SERVICE is NOT running"
        ((FAIL++))
        # Attempt to restart
        systemctl restart $SERVICE
        log " Attempted restart of $SERVICE"
    fi
}

check_port() {
    PORT=$1
    NAME=$2
    if ss -tlnp | grep -q ":$PORT"; then
        log " PASS: Port $PORT ($NAME) is open"
        ((PASS++))
    else
        log " FAIL: Port $PORT ($NAME) is NOT open"
        ((FAIL++))
    fi
}

check_disk() {
    THRESHOLD=80
    USAGE=$(df / | awk 'NR==2 {print $5}' | tr -d '%')
    if [ $USAGE -lt $THRESHOLD ]; then
        log " PASS: Disk usage is ${USAGE}% (below ${THRESHOLD}% threshold)"
        ((PASS++))
    else
        log " WARN: Disk usage is ${USAGE}% (above ${THRESHOLD}% threshold)"
        ((FAIL++))
    fi
}

check_memory() {
```

```
  FREE_MEM=$(free | awk 'NR==2{printf "%.0f", $4/$2*100}')
  if [ $FREE_MEM -gt 10 ]; then
    log " PASS: Free memory is ${FREE_MEM}%"
    ((PASS++))
  else
    log " WARN: Low memory — only ${FREE_MEM}% free"
    ((FAIL++))
  fi
}

check_http() {
  HTTP_CODE=$(curl -s -o /dev/null -w "%{http_code}" http://localhost/)
  if [ "$HTTP_CODE" == "200" ]; then
    log " PASS: HTTP response code is $HTTP_CODE"
    ((PASS++))
  else
    log " FAIL: HTTP response code is $HTTP_CODE (expected 200)"
    ((FAIL++))
  fi
}

# ---- Run All Checks ----
log "========================================"
log "Server Health Check Started"
log "========================================"

check_service "nginx"
check_service "firewalld"
check_port 80 "HTTP"
check_port 22 "SSH"
check_disk
check_memory
check_http

log "========================================"
log "Health Check Complete: PASS=$PASS | FAIL=$FAIL"
log "========================================"
echo ""
SCRIPT

# Make executable
chmod +x /usr/local/bin/server_health_check.sh
```

# Run the script manually
/usr/local/bin/server_health_check.sh

# View the log output
cat /var/log/health_check.log

**Output:**

```
[root@ip-192-168-1-235 ~]# cat > /usr/local/bin/server_health_check.sh << 'SCRIPT'
#!/bin/bash
> # ==========================================
# Server Health Check Script
# Linux Capstone Project 2
# ==========================================

LOGFILE="/var/log/health_check.log"
TIMESTAMP=$(date '+%Y-%m-%d %H:%M:%S')
PASS=0
FAIL=0

log() {
    echo "[$TIMESTAMP] $1" | tee -a $LOGFILE
}

check_service() {
    SERVICE=$1
    if systemctl is-active --quiet $SERVICE; then
        log " PASS: $SERVICE is running"
        ((PASS++))
    else
        log " FAIL: $SERVICE is NOT running"
        ((FAIL++))
        # Attempt to restart
        systemctl restart $SERVICE
        log " Attempted restart of $SERVICE"
    fi
}

check_port() {
    PORT=$1
    NAME=$2
    if ss -tlnp | grep -q ":$PORT"; then
        log " PASS: Port $PORT ($NAME) is open"
        ((PASS++))
    else
```

```
    else
        log " FAIL: Port $PORT ($NAME) is NOT open"
        ((FAIL++))
    fi
> }
>
> check_disk() {
    THRESHOLD=80
    USAGE=$(df / | awk 'NR==2 {print $5}' | tr -d '%')
    if [ $USAGE -lt $THRESHOLD ]; then
        log " PASS: Disk usage is ${USAGE}% (below ${THRESHOLD}% threshold)"
        ((PASS++))
    else
        log " WARN: Disk usage is ${USAGE}% (above ${THRESHOLD}% threshold)"
        ((FAIL++))
    fi
}

check_memory() {
    FREE_MEM=$(free | awk 'NR==2{printf "%.0f", $4/$2*100}')
    if [ $FREE_MEM -gt 10 ]; then
        log " PASS: Free memory is ${FREE_MEM}%"
        ((PASS++))
    else
        log " WARN: Low memory — only ${FREE_MEM}% free"
        ((FAIL++))
    fi
}

check_http() {
    HTTP_CODE=$(curl -s -o /dev/null -w "%{http_code}" http://localhost/)
    if [ "$HTTP_CODE" == "200" ]; then
        log " PASS: HTTP response code is $HTTP_CODE"
        ((PASS++))
    else
        log " FAIL: HTTP response code is $HTTP_CODE (expected 200)"
        ((FAIL++))
```

```
    fi
}

# ---- Run All Checks ----
log "======================================="
log "Server Health Check Started"
echo ""=======================================$FAIL"
>
> SCRIPT
[root@ip-192-168-1-235 ~]# chmod +x /usr/local/bin/server_health_check.sh
[root@ip-192-168-1-235 ~]# /usr/local/bin/server_health_check.sh
[2026-11-10 10:11:40] =======================================
[2026-11-10 10:11:40] Server Health Check Started
[2026-11-10 10:11:40] =======================================
[2026-11-10 10:11:40]  FAIL: nginx is NOT running
Job for nginx.service failed because the control process exited with error code.
See "systemctl status nginx.service" and "journalctl -xeu nginx.service" for details.
[2026-11-10 10:11:40]  Attempted restart of nginx
[2026-11-10 10:11:40]  PASS: firewalld is running
[2026-11-10 10:11:40]  PASS: Port 80 (HTTP) is open
[2026-11-10 10:11:40]  PASS: Port 22 (SSH) is open
[2026-11-10 10:11:40]  PASS: Disk usage is 22% (below 80% threshold)
[2026-11-10 10:11:40]  PASS: Free memory is 53%
[2026-11-10 10:11:40]  FAIL: HTTP response code is 403 (expected 200)
[2026-11-10 10:11:40] =======================================
[2026-11-10 10:11:40] Health Check Complete: PASS=5 | FAIL=2
[2026-11-10 10:11:40] =======================================
 fi
/usr/local/bin/server_health_check.sh: line 94: syntax error near unexpected token `}'
/usr/local/bin/server_health_check.sh: line 94: `}'
[root@ip-192-168-1-235 ~]# cat /var/log/health_check.log
[2026-11-10 10:11:40] =======================================
[2026-11-10 10:11:40] Server Health Check Started
[2026-11-10 10:11:40] =======================================
[2026-11-10 10:11:40]  FAIL: nginx is NOT running
[2026-11-10 10:11:40]  Attempted restart of nginx
[2026-11-10 10:11:40]  PASS: firewalld is running
```

```
[2026-11-10 10:11:40]  PASS: firewalld is running
[2026-11-10 10:11:40]  PASS: Port 80 (HTTP) is open
[2026-11-10 10:11:40]  PASS: Port 22 (SSH) is open
[2026-11-10 10:11:40]  PASS: Disk usage is 22% (below 80% threshold)
[2026-11-10 10:11:40]  PASS: Free memory is 53%
[2026-11-10 10:11:40]  FAIL: HTTP response code is 403 (expected 200)
[2026-11-10 10:11:40] =======================================
[2026-11-10 10:11:40] Health Check Complete: PASS=5 | FAIL=2
[2026-11-10 10:11:40] =======================================
[root@ip-192-168-1-235 ~]#
```

**Step 11: Schedule the health check with crontab**

***Commands:***

# Add to crontab — run every 5 minutes
crontab -e
# Add this line:
# */5 * * * * /usr/local/bin/server_health_check.sh

# Verify crontab
crontab -l

# Wait 5 minutes then check if it ran
tail -30 /var/log/health_check.log

**Output**:

```
[root@ip-192-168-1-235 ~]# crontab -e
no crontab for root - using an empty one
crontab: installing new crontab
[root@ip-192-168-1-235 ~]# crontab -l
*/5 * * * * /usr/local/bin/server_health_check.sh
```

```
[root@ip-192-168-1-235 ~]# tail -30 /var/log/health_check.log
[2026-11-10 10:11:40] ========================================
[2026-11-10 10:11:40] Server Health Check Started
[2026-11-10 10:11:40] ========================================
[2026-11-10 10:11:40]  FAIL: nginx is NOT running
[2026-11-10 10:11:40]  Attempted restart of nginx
[2026-11-10 10:11:40]  PASS: firewalld is running
[2026-11-10 10:11:40]  PASS: Port 80 (HTTP) is open
[2026-11-10 10:11:40]  PASS: Port 22 (SSH) is open
[2026-11-10 10:11:40]  PASS: Disk usage is 22% (below 80% threshold)
[2026-11-10 10:11:40]  PASS: Free memory is 53%
[2026-11-10 10:11:40]  FAIL: HTTP response code is 403 (expected 200)
[2026-11-10 10:11:40] ========================================
[2026-11-10 10:11:40] Health Check Complete: PASS=5 | FAIL=2
[2026-11-10 10:11:40] ========================================
```

**Step 12: Write an if-else script for Nginx management**

***Commands:***

```
cat > /usr/local/bin/manage_nginx.sh << 'SCRIPT'
#!/bin/bash

# Nginx Service Management Script

ACTION=$1

if [ -z "$ACTION" ]; then
  echo "Usage: $0 {start|stop|restart|status}"
  exit 1
fi

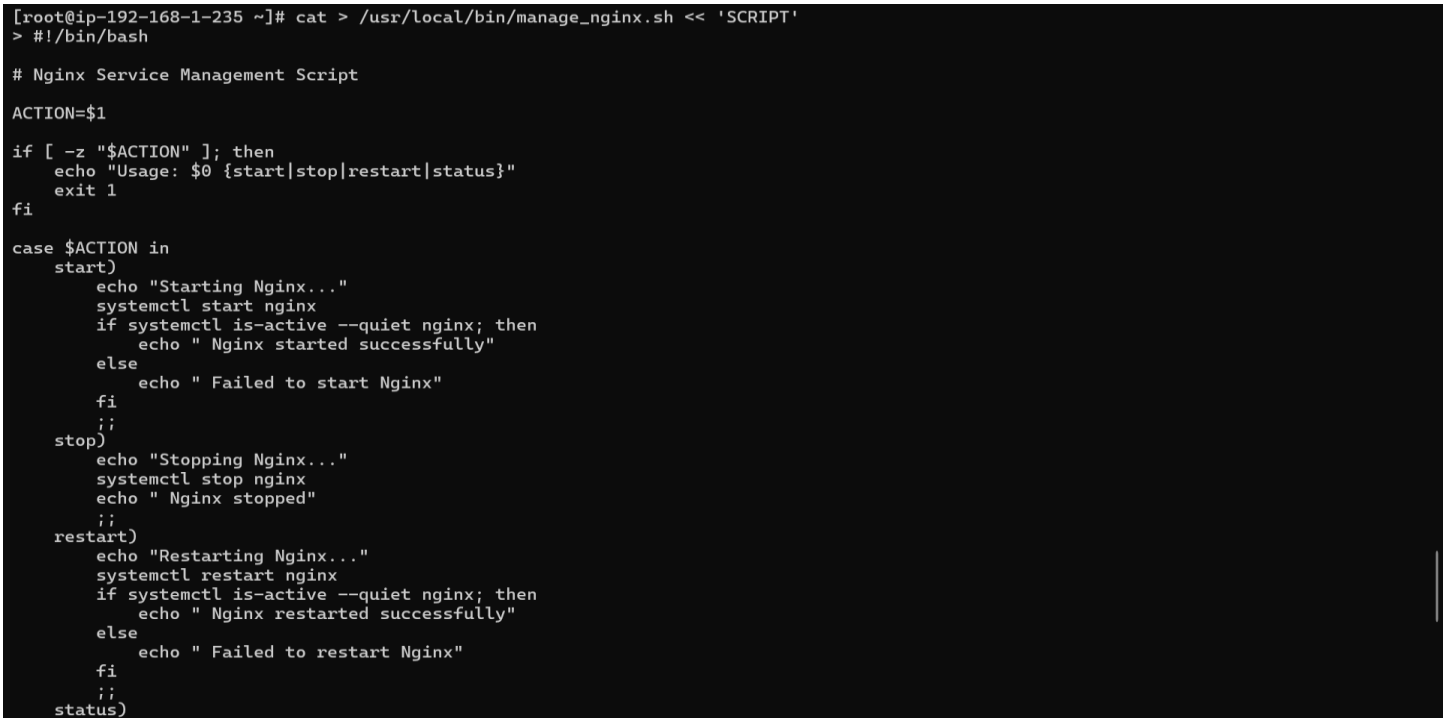
case $ACTION in
  start)
    echo "Starting Nginx..."
    systemctl start nginx
    if systemctl is-active --quiet nginx; then
      echo " Nginx started successfully"
    else
      echo " Failed to start Nginx"
    fi
    ;;
  stop)
    echo "Stopping Nginx..."
    systemctl stop nginx
    echo " Nginx stopped"
    ;;
  restart)
    echo "Restarting Nginx..."
    systemctl restart nginx
    if systemctl is-active --quiet nginx; then
```

```
        echo " Nginx restarted successfully"
    else
        echo " Failed to restart Nginx"
    fi
    ;;
  status)
    systemctl status nginx
    ;;
  *)
    echo " Invalid action: $ACTION"
    echo "Usage: $0 {start|stop|restart|status}"
    exit 1
    ;;
esac
SCRIPT

chmod +x /usr/local/bin/manage_nginx.sh

# Test all options
/usr/local/bin/manage_nginx.sh status
/usr/local/bin/manage_nginx.sh stop
/usr/local/bin/manage_nginx.sh start
/usr/local/bin/manage_nginx.sh restart
/usr/local/bin/manage_nginx.sh invalid
```

***Output:***



```
[root@ip-192-168-1-235 ~]# cat > /usr/local/bin/manage_nginx.sh << 'SCRIPT'
> #!/bin/bash

# Nginx Service Management Script

ACTION=$1

if [ -z "$ACTION" ]; then
    echo "Usage: $0 {start|stop|restart|status}"
    exit 1
fi

case $ACTION in
    start)
        echo "Starting Nginx..."
        systemctl start nginx
        if systemctl is-active --quiet nginx; then
            echo " Nginx started successfully"
        else
            echo " Failed to start Nginx"
        fi
        ;;
    stop)
        echo "Stopping Nginx..."
        systemctl stop nginx
        echo " Nginx stopped"
        ;;
    restart)
        echo "Restarting Nginx..."
        systemctl restart nginx
        if systemctl is-active --quiet nginx; then
            echo " Nginx restarted successfully"
        else
            echo " Failed to restart Nginx"
        fi
        ;;
    status)
```

```
    status)
        systemctl status nginx
        ;;
>   *)
>       echo " Invalid action: $ACTION"
>       echo "Usage: $0 {start|stop|restart|status}"
>       exit 1
>       ;;
>
> esac
>
> SCRIPT
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# chmod +x /usr/local/bin/manage_nginx.sh
[root@ip-192-168-1-235 ~]# /usr/local/bin/manage_nginx.sh status
× nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
     Active: failed (Result: exit-code) since Tue 2026-11-10 12:05:04 UTC; 2min 12s ago
    Process: 2959 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 2960 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 2961 ExecStart=/usr/sbin/nginx (code=exited, status=1/FAILURE)
        CPU: 40ms

Nov 10 12:05:02 ip-192-168-1-235.ap-south-1.compute.internal nginx[2961]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address al>
Nov 10 12:05:02 ip-192-168-1-235.ap-south-1.compute.internal nginx[2961]: nginx: [emerg] bind() to [::]:80 failed (98: Address alrea>
Nov 10 12:05:03 ip-192-168-1-235.ap-south-1.compute.internal nginx[2961]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address al>
Nov 10 12:05:03 ip-192-168-1-235.ap-south-1.compute.internal nginx[2961]: nginx: [emerg] bind() to [::]:80 failed (98: Address alrea>
Nov 10 12:05:03 ip-192-168-1-235.ap-south-1.compute.internal nginx[2961]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address al>
Nov 10 12:05:03 ip-192-168-1-235.ap-south-1.compute.internal nginx[2961]: nginx: [emerg] bind() to [::]:80 failed (98: Address alrea>
Nov 10 12:05:04 ip-192-168-1-235.ap-south-1.compute.internal nginx[2961]: nginx: [emerg] still could not bind()
Nov 10 12:05:04 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: nginx.service: Control process exited, code=exited, status=>
Nov 10 12:05:04 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: nginx.service: Failed with result 'exit-code'.
Nov 10 12:05:04 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Failed to start nginx.service - The nginx HTTP and reverse >

[root@ip-192-168-1-235 ~]# /usr/local/bin/manage_nginx.sh stop
Stopping Nginx...
 Nginx stopped
```

```
[root@ip-192-168-1-235 ~]# /usr/local/bin/manage_nginx.sh start
Starting Nginx...
Job for nginx.service failed because the control process exited with error code.
See "systemctl status nginx.service" and "journalctl -xeu nginx.service" for details.
 Failed to start Nginx
[root@ip-192-168-1-235 ~]# /usr/local/bin/manage_nginx.sh restart
Restarting Nginx...
Job for nginx.service failed because the control process exited with error code.
See "systemctl status nginx.service" and "journalctl -xeu nginx.service" for details.
 Failed to restart Nginx
[root@ip-192-168-1-235 ~]# /usr/local/bin/manage_nginx.sh invalid
[root@ip-192-168-1-235 ~]#
```

**Step 13: Final system state verification**
**Commands:**

```
# Final check of everything
echo "=== NGINX STATUS ==="
systemctl status nginx --no-pager

echo "=== FIREWALL RULES ==="
firewall-cmd --list-all

echo "=== OPEN PORTS ==="
ss -tlnp

echo "=== DISK USAGE ==="
df -h

echo "=== MEMORY ==="
free -h
```

echo "=== CRONTAB ==="
crontab -l

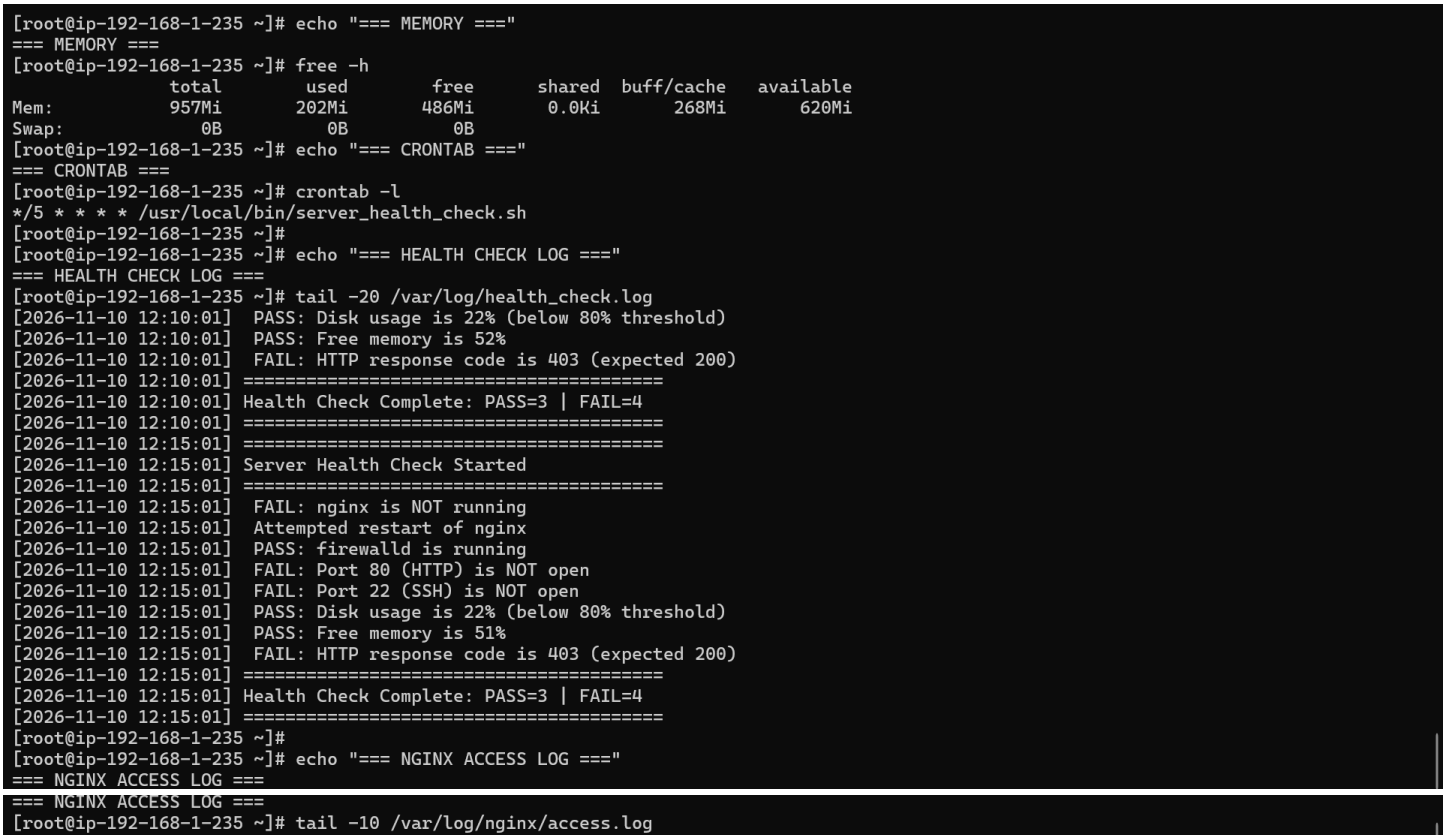
echo "=== HEALTH CHECK LOG ==="
tail -20 /var/log/health_check.log

echo "=== NGINX ACCESS LOG ==="
tail -10 /var/log/nginx/access.log

**Output:**

```
[root@ip-192-168-1-235 ~]# echo "=== NGINX STATUS ==="
=== NGINX STATUS ===
[root@ip-192-168-1-235 ~]# systemctl status nginx --no-pager
× nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
     Active: failed (Result: exit-code) since Tue 2026-11-10 12:10:04 UTC; 4min 40s ago
    Process: 3345 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 3346 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 3347 ExecStart=/usr/sbin/nginx (code=exited, status=1/FAILURE)
        CPU: 39ms

Nov 10 12:10:02 ip-192-168-1-235.ap-south-1.compute.internal nginx[3347]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Addr…n use)
Nov 10 12:10:02 ip-192-168-1-235.ap-south-1.compute.internal nginx[3347]: nginx: [emerg] bind() to [::]:80 failed (98: Address…n use)
Nov 10 12:10:03 ip-192-168-1-235.ap-south-1.compute.internal nginx[3347]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Addr…n use)
Nov 10 12:10:03 ip-192-168-1-235.ap-south-1.compute.internal nginx[3347]: nginx: [emerg] bind() to [::]:80 failed (98: Address…n use)
Nov 10 12:10:03 ip-192-168-1-235.ap-south-1.compute.internal nginx[3347]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Addr…n use)
Nov 10 12:10:03 ip-192-168-1-235.ap-south-1.compute.internal nginx[3347]: nginx: [emerg] bind() to [::]:80 failed (98: Address…n use)
Nov 10 12:10:04 ip-192-168-1-235.ap-south-1.compute.internal nginx[3347]: nginx: [emerg] still could not bind()
Nov 10 12:10:04 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: nginx.service: Control process exited, code=exited, s…AILURE
Nov 10 12:10:04 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: nginx.service: Failed with result 'exit-code'.
Nov 10 12:10:04 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Failed to start nginx.service - The nginx HTTP and re…erver.
Hint: Some lines were ellipsized, use -l to show in full.
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# echo "=== FIREWALL RULES ==="
=== FIREWALL RULES ===
[root@ip-192-168-1-235 ~]# firewall-cmd --list-all
public
  target: default
  icmp-block-inversion: no
  interfaces:
  sources:
  services: dhcpv6-client http mdns ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@ip-192-168-1-235 ~]# echo "=== OPEN PORTS ==="
=== OPEN PORTS ===
[root@ip-192-168-1-235 ~]# ss -tlnp
State            Recv-Q            Send-Q                       Local Address:Port                       Peer Address:Port
Process
LISTEN           0                 128                                0.0.0.0:22                              0.0.0.0:*
 users:(("sshd",pid=2680,fd=7))
LISTEN           0                 511                                   *:80                                    *:*
 users:(("httpd",pid=2331,fd=4),("httpd",pid=2330,fd=4),("httpd",pid=2323,fd=4),("httpd",pid=2312,fd=4))
LISTEN           0                 128                                [::]:22                                 [::]:*
 users:(("sshd",pid=2680,fd=8))
[root@ip-192-168-1-235 ~]# echo "=== DISK USAGE ==="
=== DISK USAGE ===
[root@ip-192-168-1-235 ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           479M     0  479M   0% /dev/shm
tmpfs           192M  460K  192M   1% /run
/dev/xvda1      8.0G  1.7G  6.3G  22% /
tmpfs           479M     0  479M   0% /tmp
/dev/xvda128     10M  1.3M  8.7M  13% /boot/efi
tmpfs            96M     0   96M   0% /run/user/1000
```



```
[root@ip-192-168-1-235 ~]# echo "=== MEMORY ==="
=== MEMORY ===
[root@ip-192-168-1-235 ~]# free -h
               total        used        free      shared  buff/cache   available
Mem:           957Mi       202Mi       486Mi       0.0Ki       268Mi       620Mi
Swap:             0B          0B          0B
[root@ip-192-168-1-235 ~]# echo "=== CRONTAB ==="
=== CRONTAB ===
[root@ip-192-168-1-235 ~]# crontab -l
*/5 * * * * /usr/local/bin/server_health_check.sh
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# echo "=== HEALTH CHECK LOG ==="
=== HEALTH CHECK LOG ===
[root@ip-192-168-1-235 ~]# tail -20 /var/log/health_check.log
[2026-11-10 12:10:01]  PASS: Disk usage is 22% (below 80% threshold)
[2026-11-10 12:10:01]  PASS: Free memory is 52%
[2026-11-10 12:10:01]  FAIL: HTTP response code is 403 (expected 200)
[2026-11-10 12:10:01] ======================================
[2026-11-10 12:10:01] Health Check Complete: PASS=3 | FAIL=4
[2026-11-10 12:10:01] ======================================
[2026-11-10 12:15:01] ======================================
[2026-11-10 12:15:01] Server Health Check Started
[2026-11-10 12:15:01] ======================================
[2026-11-10 12:15:01]  FAIL: nginx is NOT running
[2026-11-10 12:15:01]  Attempted restart of nginx
[2026-11-10 12:15:01]  PASS: firewalld is running
[2026-11-10 12:15:01]  FAIL: Port 80 (HTTP) is NOT open
[2026-11-10 12:15:01]  FAIL: Port 22 (SSH) is NOT open
[2026-11-10 12:15:01]  PASS: Disk usage is 22% (below 80% threshold)
[2026-11-10 12:15:01]  PASS: Free memory is 51%
[2026-11-10 12:15:01]  FAIL: HTTP response code is 403 (expected 200)
[2026-11-10 12:15:01] ======================================
[2026-11-10 12:15:01] Health Check Complete: PASS=3 | FAIL=4
[2026-11-10 12:15:01] ======================================
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# echo "=== NGINX ACCESS LOG ==="
=== NGINX ACCESS LOG ===
```



```
=== NGINX ACCESS LOG ===
[root@ip-192-168-1-235 ~]# tail -10 /var/log/nginx/access.log
```

THANK YOU !!